Arjav Poudel

(+44) 7479917110 | poudelarjav@gmail.com | arjavpoudel.github.io | linkedin.com/in/arjavpoudel | github.com/arjavpoudel

Education

University of Edinburgh — Edinburgh, UK
BSc (Hons) Artificial Intelligence — *Sep. 2022 – June 2026*

- Degree Classification: First Class Honours (1st)
- Relevant Courses: Machine Learning (82%), Informatics Large Practical (86%),Computational Cognitive Science (76%), Computational Neuroscience (83%) Natural Language Processing (70%), Machine Learning Practical (78%)

Grove Academy — Dundee, UK
Scottish Highers / Advanced Highers — *Sep. 2020 – June 2022*

- 5 A1s at Higher; A1s in Advanced Higher Maths and Physics; Scottish Baccalaureate in Science - Distinction

Experience

ML Engineer Summer Intern — June 2024 – Aug. 2024
UK Atomic Energy Authority (UKAEA) — *Culham, UK*

- Developed a novel ML solution for experimental validation in fusion technology, focusing on emerging physics-informed approaches
- Engineered and deployed 'Icarus-Fusion', an open-source machine learning tool that aids experimental validation processes, on PyPI
- Implemented and optimised four model architectures (MLPs, CNNs, GANs, PINNs) to predict discrepancies between simulations and ground-truth experimental data
- Explored the application of Physics-Informed Neural Networks (PINNs) in fusion-based validation
- Presented project at internal research symposium

Head of Computer Vision — Sep. 2024 – Sep. 2025
HumanEd – University of Edinburgh Society — *Edinburgh, UK*

- Led development of an advanced bio-mimetic robotic hand that uses computer vision and deep reinforcement learning to autonomously solve a Rubik's cube
- Led the Computer Vision team in leveraging image classification, object detection, and scene segmentation to deploy a generalised infrastructure capable of solving cube puzzles without specialised sensors or modifications
- Built upon prior work by OpenAI, which relied on sensor-embedded "tricked cubes," enhancing and refining the existing approach to push the boundaries of what's possible without specialised hardware

Projects

Honours Project – Diffusion Modelling of Neural Dynamics | *PyTorch* — June 2025 – April 2026

- Discovered and mathematically proved a hidden flaw in a major class of AI models.
- Proved the cause of this flaw using a well-known statistical model, then confirmed it experimentally across four different test scenarios, including complex, 50 dimensional cases.
- Proposed and implemented a Centred Architecture that resolves the failure mode, achieving accurate results where the standard approach broke down completely.

Nepali Handwritten Character Recognition System | *PyTorch, CNNs*

- Developed a model using convolutional neural networks to recognise handwritten Nepali characters, achieving over 97% accuracy on a limited training dataset
- Built a software tool with a real-time handwriting recognition algorithm aimed at supporting language learning for underserved Nepali communities in the UK

Technical Skills

Languages: Python, C, Haskell, Java, SQL, Swift, JavaScript, MIPS Assembly
Frameworks: PyTorch, OpenCV, Pandas, NumPy, FastAPI, Django, Scikit-learn
Technologies: Formal Logic, Automated Theorem Proving, Formal Verification (Isabelle/HOL), Computer Vision, Git, UNIX, Scientific Computing, Physics-Informed ML, CI/CD